Q:5 With neat sketches, briefly describe Neighborhood, Adjacency, Connectivity, Paths,Regions and Boundaries of an image pixel.

Neighborhood

- $N_4(p)$: 4 neighbors (Up, Down, Left, Right).
- $N_D(p)$: 4 Diagonal neighbors.
- $N_8(p)$: $N_4(p) \cup N_D(p)$ (All 8 neighbors).

Adjacency

Pixels p, q with values from set V are adjacent if:

- **4-Adjacency:** $q \in N_4(p)$.
- **8-Adjacency:** $q \in N_8(p)$.
- **m-Adjacency:** $q \in N_4(p)$ OR ($q \in N_D(p)$ AND $N_4(p) \cap N_4(q)$ has no pixels from V).

Connectivity

3. Connectivity

Two pixels p, q are **connected** if there is a **path** between them consisting only of pixels from a set S.

- Types: 4, 8, or m-connectivity.

4. Path

A sequence of pixels $p_1, p_2, \ldots, p_n$ where p_i is **adjacent** to p_{i-1}.

- **Closed Path:** If $p_1 = p_n$.

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5. Region and Boundary

- **Region (R):** A connected set of pixels.
- **Boundary (Border):** Pixels in R that have at least one neighbor **not** in R.